



The art of flirting: What are the traits that make it effective?

Menelaos Apostolou*, Christoforos Christoforou

Department of Social Sciences, University of Nicosia, 46 Makedonitissas Ave., Nicosia, 1700, Cyprus



ARTICLE INFO

Keywords:

Flirting
Flirting effectiveness
Mate preferences
Mating
Sex-differences

ABSTRACT

Flirting is an essential aspect of human interaction and key for the formation of intimate relationships. In the current research, we aimed to identify the traits that turn it more effective. In particular, in Study 1 we used open-ended questionnaires in a sample of 487 Greek-speaking participants, and identified 47 traits that make flirting effective. In Study 2, we asked 808 Greek-speaking participants to rate how effective each trait would be on them. Using principal components analysis, we classified these traits into nine broader factors. Having a good non-verbal behavior, being intelligent and having a gentle approach, were rated as the most important factors. Sex difference were found for most of the factors. For example, women rated gentle approach as more effective on them, while men rated good looks as more effective. Last but not least, older participants rated factors, such as the "Gentle approach," to be more effective on them.

1. Introduction

People are disposed to form intimate relationships, and in order to do so they need to persuade prospective mates to make a casual or a long-term relationship with them. The usual way to achieve this goal is through flirting, a universal and essential aspect of human interaction (Eibl-Eibesfeldt & Hass, 1967; Luscombe, 2008). Flirting could be used to serve purposes other than mating (Henningesen, 2004; Scheflen, 1965). In particular, people may flirt because it constitutes an enjoyable form of interaction (Guerrero, Andersen & Afifi, 2001), in order to build their own self-esteem (Loe, 1996) and in order to accomplish an instrumental goal, such as having someone to buy them a drink (Yelvington, 1996). However, the primary function of flirting is to establish a long- or a short-term intimate relationship, and in the present research, we will focus on flirting in the context of mating. Thus, for our purposes, flirting refers to covert signals sent between individuals to indicate sexual interest and begin courtship (White, Lorenz, Perilloux & Lee, 2018). The current research attempted to address the question of what traits make flirting effective, and whether the effectiveness of each trait depends on the sex of the target of flirting. These traits could be better understood within an evolutionary theoretical context that will be discussed next.

2. Traits that make flirting effective

2.1. Mate preferences

Flirting would be successful, if it manages to convey the message that the individual who engages in it has the qualities that the other party desires (Wade & Feldman, 2016). Accordingly, in order predict which traits would make flirting more effective, we need to know what people look for in a partner. People do not choose mates randomly, but they have well-defined preferences to guide their choices (Buss, 2017). More specifically, mating has tremendous evolutionary importance, because those who fail to gain access to mates do not pass their genetic material to future generations. The evolutionary importance of mating, translates into very strong selection pressures shaping the human mind to divert effort in finding mates. Prospective mates vary in their qualities and some qualities are beneficial, while others are harmful to the fitness of mate-seekers. For example, people who are not healthy, carry many harmful genetic mutations, or are old, are unlikely to make good mates, because they may soon die or become incapacitated. On the other hand, people who are healthy, do not carry many harmful genetic mutations, or are young, have better prospects as mates.

Accordingly, selection forces have not shaped the human mind to divert effort in finding mates indiscriminately but selectively. That is, people have evolved preferences that enable them to divert their mating effort in getting mates with fitness-increasing traits, and avoid mates with fitness-decreasing traits. Research efforts in the area have found that people tend to prefer good character, good looks, intelligence,

* Corresponding author.

E-mail address: m.apostolou@gmail.com (M. Apostolou).

<https://doi.org/10.1016/j.paid.2020.109866>

Received 4 January 2020; Received in revised form 21 January 2020; Accepted 24 January 2020

0191-8869/ © 2020 Elsevier Ltd. All rights reserved.

similarity, capacity to generate resources, control of wealth and willingness to commit to a relationship (Buss, 2017; Buss & Barnes, 1986; Buss & Schmitt, 2019; Thomas et al., 2019). On this basis, we could predict that the elements which would make flirting more effective, are the ones that convey the information to the target that the initiator of flirting has these qualities. For instance, good looks, effective demonstration of intelligence and interest could be traits that would make flirting effective.

2.2. Sex differences

Being able to provide the necessary conditions for a fertilized egg to develop into a baby requires specialized biological machinery, which in turn, favors the differentiation of sexes, with one sex, the female, specializing in developing this hardware. This difference in specialization, has several implications, which materialize in different selection pressures exercised on each sex that in turn, result in behavioral and physiological differences between them. One such implication is that certain traits offer different benefits to men and women (Buss, 1989). For instance, commitment to a relationship is more valuable to a woman, who needs someone to help her during her pregnancy and the months following it (Buss, 2017).

The sex differences in mate preferences would reflect on the traits that make flirting more likely to succeed; that is to say, specific traits would make flirting more successful, if the target is a woman than if the target is a man, or the other way round. Research on mate preferences has found that women ascribe more importance than men to the willingness to commit to a relationship and control resources, and men ascribe more importance than women to the looks of prospective mates (Buss, 1989; 2017; Furnham, 2009; Li, Valentine & Patel, 2010). On this basis, we predict that women would be more likely to give in to flirting, if the prospective mate indicates a strong willingness to be with them and appears to be well-off, while men would be more likely to yield to flirting, if the prospective mate is good looking.

3. Current literature

There have been several studies on identifying flirting acts (Deshotels & Forsyth, 2006; Farley 2014; Grammer, 1990, 2000; Moore, 1985; Moore & Butler, 1989; Renninger, Wade & Grammer, 2004 see Moore, 2010; for a comprehensive overview on nonverbal flirting behaviors). Some of the most common ones include smiling (Gueguen, 2008; McCormick & Jones, 1989), eye contact (Moore, 1985; Walsh & Hewitt, 1985), laughing (Grammer, 1990; Moore, 1985) and touching (Clark, Shaver & Abrahams, 1999; Moore, 1985; Willis & Briggs, 1992). However, the literature on the effectiveness of such acts is rather limited.

To begin with, Walsh and Hewitt (1985) had a female confederate to establish eye contact with a man in a bar. After such contact was established, the confederate smiled or did not smile, with the researchers examining whether the subject approached and talked to her. They found that the highest approach rate (60%), was observed when there was repeated eye contact plus smiling.

In the most comprehensive study on flirting effectiveness so far, Wade and Feldman (2016) identified 26 acts that men would use, and 14 acts that women would use in the context of flirting. They also found that, for men the more effective acts were "He asks her out," "He spends time with her," "He kisses her," and "He acts interested in her." For women, the more effective acts were "She moves closer to him," "She kisses him on the cheek," "She shows interest in him during a conversation" and "She touches him, in general." They also found significant sex differences; male participants rated flirtations from women that suggested sexual access as more effective, while women rated flirtations from men that suggested exclusivity, commitment and caring as the most effective flirtations.

White et al. (2018) employed a sample of 150 participants in the US

and asked them to list flirting behaviors, as well as to rate each behavior for its effectiveness. Overall, they identified 450 such behaviors. The ones with the highest scores included "I would be confident," "I would listen to him/her and be engaged," "I would smile at him/her," and "I would ask him/her questions about her/himself."

We were not able to locate any research, which has specifically attempted to identify the traits that make flirting effective; however, some of these traits could be indirectly inferred by the literature above. For instance, from the Wade and Feldman (2016) study, we could infer that a strong demonstration of interest would be a vital component of effective flirting. From the White et al. (2018) study, we could infer that being confident, demonstrating a strong interest and being smiley would be important components of effective flirting. Nevertheless, such attempt is constrained by the limitations of the current literature, including failure to classify acts into broader categories and using small samples. For instance, it would be difficult to say which traits make flirting effective by going through the 450 behaviors that White et al. (2018) have identified. Accordingly, the current research aimed to expand the limited literature on the effectiveness of flirting by identifying the traits that make flirting effective, classifying them in broader categories and examining whether their effectiveness is different on men and women. Finally, the existing literature was based on American samples, and to our knowledge, the current study is the first which has attempted to examine flirting in a different cultural context, namely the Greek one.

4. Study 1

The present study aimed to identify the traits that make flirting effective.

4.1. Methods

4.1.1. Participants

The study was designed in a private university in the Republic of Cyprus, was in Greek and ran online. The link of the study was promoted as a Facebook advertisement to participants who resided in Greece and in the Republic of Cyprus. The only condition for participating was to be at least 18 years old. Participants did not receive any monetary or other reward. Overall, 487 Greek-speaking individuals took part (281 women, 204 men and two participants who did not indicate their sex). The mean age of women was 24.6 ($SD = 7.1$), and the mean age of men was 28.5 ($SD = 9.3$). Finally, 50.7% of the participants were in a relationship, 37.4% were single, 9.6% were married and 2.3% were divorced.

4.1.2. Materials

The open-ended survey was designed using Google Forms and had two parts. In the first part, participants were asked the following: "People in the past may have flirted with you. Write down some of the positive things you found in their flirting that motivated you to consider giving in to their attempts." In the second part, demographic information was collected (i.e., sex, age and marital status).

4.2. Results

In order to analyze the data from the open-ended questionnaires, we employed methods developed in prior research (Apostolou, 2017; Apostolou & Panayiotou, 2019). More specifically, two independent graduate students were employed, a man and a woman, who coded and categorized responses to supraordinate categories. Similar responses were added to a supraordinate category and, when a dissimilar response arose, a new supraordinate category was created. Answers that contained multiple reasons were eliminated, as they were difficult to interpret. In addition, reasons with unclear or vague wording were also eliminated. After processing about 30% of the responses, the data for

Table 1

Frequencies and mean scores for the 47 traits which turn flirting effective from Studies 1 and 2.

Traits	Frequency	Mean (SD)	% of participants who gave scores "4" or "5"
Humor	216	4.47 (0.76)	88.8
Intelligence	117	4.40 (0.70)	87.5
Good looks	97	3.78 (0.97)	62.6
Politeness	90	4.42 (0.76)	88.5
Honesty	54	4.47 (0.82)	88
Intense interest in me	48	4.18 (0.96)	77.7
Good with words	42	3.94 (0.94)	69.1
Persistence	37	3.63 (1.16)	57.5
Respect	36	4.51 (0.75)	89.7
Good character	31	4.42 (0.81)	86.3
Self-confidence	30	4.03 (0.94)	71.6
Originality	29	3.77 (1.07)	63.2
Nice smile	28	4.26 (0.89)	82.1
Way of approach	28	4.21 (0.91)	78.4
Easygoingness	26	3.69 (1.05)	59.2
Common characteristics	25	3.70 (1.04)	57.9
Courage	24	3.97 (0.95)	71.1
Discreteness	22	3.98 (0.96)	69.6
Interesting personality	22	4.37 (0.82)	85.0
Charm	21	4.26 (0.80)	83.0
Common interests	20	3.80 (0.99)	63.1
Education	18	3.67 (1.02)	58.7
Tenderness	16	4.17 (0.90)	77.4
Pleasantness	14	4.37 (0.77)	86.4
Spontaneity	14	3.98 (0.96)	71.9
Directness	13	4.07 (0.92)	69.5
Intense gaze	13	4.08 (0.98)	73.7
Kindness	13	4.25 (0.91)	80.8
Mystery	13	3.34 (1.19)	44.4
Comfortable with his/her spending	12	2.78 (1.22)	26.4
Nice compliments	12	3.50 (1.12)	51.3
Patience	12	3.84 (1.06)	64.9
Romance	12	3.81 (1.06)	62.3
Sweetness	12	4.04 (0.98)	73.5
Boldness	11	3.96 (0.93)	70.7
Determination	11	4.07 (0.91)	73.8
Directness	11	3.95 (0.94)	73.1
Good dressing	11	3.59 (1.04)	55.6
Interest to learn things about me	11	4.14 (0.92)	78.3
Positivity	11	4.24 (0.87)	81.8
The way he/she looks at me	11	4.33 (0.83)	84.9
Cheerfulness	10	4.09 (0.91)	76.0
Maturity	10	4.24 (0.87)	81.6
Nice smell	10	4.24 (0.90)	81.4
Seriousness	10	3.78 (1.04)	62.2
She/he is not moving fast	10	3.17 (1.13)	36.4
Unusual hobbies	10	2.89 (1.18)	29.6

each coder were compared. The coders agreed to most of the supraordinate categories. For some cases without complete overlap, one of the authors was consulted, and eventually agreement was reached for 100% of the supraordinate categories. Subsequently, coders proceeded to code the remaining responses. Overall, 47 traits that make flirting effective were identified and they are listed in Table 1.

In order to get a measure of importance, we estimated the frequency that each trait was reported, and we placed them in a hierarchical order. As we can see in Table 1, the most frequently reported one by far was "humor," followed by "intelligence" and "good looks." In the middle of the hierarchy were traits, such as "way of approach," "easygoingness" and "courage," while at the bottom of the hierarchy were traits, such as "nice smell" and "unusual hobbies."

5. Study 2

The present study aimed to classify the traits identified in Study 1 to broader categories, and to assess their importance and their contingencies.

5.1. Methods

5.1.1. Participants

The study was designed and ran in a private university in the Republic of Cyprus. The way of recruiting participants was the same as in Study 1. Overall, 808 Greek-speaking individuals took part (415 women, 393 men and two participants who did not indicate their sex). The mean age of women was 34.7 ($SD = 11.3$) and the mean age of men was 38.2 ($SD = 12.1$). Moreover, 32.6% of the participants were single, 31.6% were married, 22.3% were in a relationship, 14.7% were divorced and 1.7% were widowed.

5.1.2. Materials

The study was designed using Google forms, was in Greek, and ran online. The survey had two parts. In the first part, participants were given the following scenario: "Please consider the scenario that someone is flirting with you. Rate how likely each of the following will make you give in to his/her flirting attempts." Participants were subsequently asked to rate each of the 47 traits identified in Study 2. Participants' scores were recorded in a five-point Likert scale: 1 – highly unlikely, 5 – very likely. In the second part, participants' age, sex and

Table 2
The extracted factors and the respective factor loadings in Study 2.

Factors	Factor loadings
Gentle approach	
She/he is not moving fast	0.635
Seriousness	0.634
Respect	0.514
Discreteness	0.494
Maturity	0.469
Politeness	0.408
Good character	0.383
Patience	0.380
Kindness	0.320
Honesty	0.286
Good looks	
Good looks	0.715
Good dressing	0.658
Comfortable with his/her spending	0.628
Charm	0.320
Courage and determination	
Courage	−0.723
Determination	−0.679
Boldness	−0.676
Directness	−0.590
Straightness	−0.578
Self-confidence	−0.494
Persistence	−0.395
Intelligence	
Intelligence	−0.708
Education	−0.605
Interesting personality	−0.595
Good with words	−0.527
Humor	−0.506
Cheerfulness	
Cheerfulness	−0.502
Easygoingness	−0.478
Spontaneity	−0.450
Pleasantness	−0.416
Positivity	−0.365
Non-verbal behaviors	
The way he/she looks at me	−0.731
Intense gaze	−0.667
Nice smile	−0.563
Nice smell	−0.453
Way of approach	−0.280
Mystery	
Mystery	0.634
Unusual hobbies	0.600
Originality	0.518
Common characteristics	
Common characteristics	−0.876
Common interests	−0.858
Strong interest and romance	
Romance	−0.696
Tenderness	−0.650
Intense interest in me	−0.643
Sweetness	−0.560
Interest to learn things about me	−0.495
Nice compliments	−0.485

marital status were recorded.

5.2. Results

In order to classify the traits that made flirting effective into broader factors, principal components analysis with direct oblimin as a rotation method was used Table 2. The KMO statistic was 0.94, indicating that our sample was very good for principal components analysis to be performed. On the basis of the Kaiser criterion (eigenvalue > 1), nine factors were extracted. The internal consistency scores ranged from 0.61 to 0.84 with a mean of 0.74. In order to examine whether the extracted factors could be classified into broader domains, second order principal components analysis was performed. In particular, nine new variables were created by averaging the scores of the traits that

composed each factor. Principal components analysis with direct oblimin as a rotation method was applied to these variables. The results indicated a one factor solution, suggesting that these factors could not be classified to broader domains. Moreover, we have also performed Pearson's product moment correlation on the nine variables. All correlations were significant, while the correlation coefficients were positive and ranged from small to moderate in strength. These correlations, along with the failure to produce a second order factor structure, suggested that there is one domain of flirting effectiveness with nine facets.

The first factor to emerge was the “Gentle approach,” where participants indicated that they would become more susceptible to flirting, if the initiators were not moving fast, were discrete, mature, respectful and polite. The second factor to emerge was the “Good looks,” where participants indicated that they would give in to flirting, if the initiators were good looking, well-dressed and charming. In the “Courage and determination,” traits, such as courage, determination, confidence and persistence, made flirting more effective. In the “Intelligence,” intelligence and traits which connoted intelligence, such as education and humor, made people more likely to give in. Furthermore, the “Cheerfulness” factor indicated that cheerfulness, easygoingness, spontaneity and positivity were traits that turned flirting effective.

As indicated by the “Non-verbal behaviors” factor, making good eye contact and having a good smell would make flirting more effective. Moreover, as indicated by the “Mystery” factor, giving the impression of a mystery, having unusual hobbies and originality were elements of effective flirting. In addition, the “Common characteristics” factor indicated that having things in common increases flirting effectiveness. Finally, its effectiveness would increase if flirting had elements of “Strong interest and romance.”

In order to examine which factors were rated as more important, we estimated the means and the standard deviations for each one and we put them in a hierarchical order, by placing first the one with the highest mean and last the one with the lowest mean. As we can see from Table 1, the “Non-verbal behaviors,” the “Intelligence,” the “Gentle approach” and the “Cheerfulness” were found at the top of the hierarchy. The “Good looks” and the “Mystery” factors were at the bottom of the hierarchy.

In order to identify sex and age effects, we performed a series of MANCOVAs. More specifically, the reasons composing each factor were entered as the dependent variables, the sex was entered as the categorical independent variable, and the age was entered as the continuous independent variable. Overall, nine tests were performed, and the results are presented in Table 3. As multiple tests were applied, Bonferroni correction for alpha inflation could be used, reducing the alpha to 0.005 (0.05/9). Accordingly, the reader may consider any results with a *p*-value higher than 0.005 as not significant.

With respect to age, significant effects were found for four factors. In particular, for the “Gentle approach,” the “Cheerfulness” and the “Good looks,” the regression coefficients were positive, indicating that older participants gave higher scores. In the “Strong interest and romance” factor, the regression coefficient was positive for the “tenderness” trait, while the regression coefficients were negative for the “interest to learn things about me” and the “nice compliments” traits. As indicated by the effect size, the largest age effect was for the “Strong interest and romance,” followed by the “Gentle approach” and the “Cheerfulness.”

Significant sex differences were found for most factors. As indicated by the effect size, the largest difference was for the “Non-verbal behaviors,” where women gave higher scores than men. The second largest difference was for the “Good looks” factor. In particular, for the “good looks” trait men gave significantly higher scores ($M = 4.03$, $SD = 0.88$) than women ($M = 3.56$, $SD = 0.98$) (Cohen's $d = 0.51$), while for the “comfortable with his/her spending” trait women gave significantly higher scores ($M = 2.86$, $SD = 1.17$) than men ($M = 2.66$, $SD = 1.25$) (Cohen's $d = 0.17$). For the “Gentle approach,” the “Intelligence,” and

Table 3
The hierarchy of factors, and sex and age effects in Study 2.

Factors	Overall Mean (SD)	Women Mean (SD)	Men Mean (SD)	p-value	η_p^2	Age p-value	η_p^2
Non-verbal behaviors	4.22 (0.62)	4.30 (0.58)	4.14 (0.64)	< 0.001	0.102	0.560	0.006
Intelligence	4.17 (0.60)	4.28 (0.54)	4.04 (0.63)	< 0.001	0.048	0.276	0.009
Gentle approach	4.10 (0.59)	4.21 (0.54)	3.98 (0.61)	< 0.001	0.093	< 0.001	0.050
Cheerfulness	4.07 (0.66)	4.05 (0.65)	4.09 (0.66)	.113	0.013	< 0.001	0.033
Strong interest and romance	3.97 (0.68)	3.98 (0.68)	3.96 (0.68)	< 0.001	0.063	< 0.001	0.053
Courage and determination	3.95 (0.66)	4.04 (0.63)	3.85 (0.68)	< 0.001	0.040	0.093	0.017
Common characteristics	3.75 (0.89)	3.76 (0.86)	3.74 (0.96)	.987	0.000	0.413	0.003
Good looks	3.60 (0.72)	3.55 (0.75)	3.65 (0.67)	< 0.001	0.099	0.004	0.022
Mystery	3.33 (0.86)	3.34 (0.83)	3.32 (0.88)	< 0.001	0.040	0.035	0.012

the “Courage and determination” factors women gave significantly higher scores than men. The effect size indicated that these differences were moderate.

In addition, a moderate in size sex difference emerged for the “Strong interest and romance,” where women gave significantly higher scores than men for the “intense interest for me” (women: $M = 4.27$, $SD = 0.89$; men $M = 4.10$, $SD = 1.00$; Cohen's $d = 0.18$) and for the “interest to learn things about me” (women: $M = 4.24$, $SD = 0.91$; men $M = 4.07$, $SD = 0.93$; Cohen's $d = 0.19$), while men gave significantly higher scores ($M = 4.19$, $SD = 0.90$) than women ($M = 3.88$, $SD = 1.01$) (Cohen's $d = 0.32$) for the “sweetness” trait. In the “Mystery” factor, where the sex difference was moderate to small, for the “unusual interests” trait, men gave significantly higher scores ($M = 2.95$, $SD = 1.19$) than women ($M = 2.80$, $SD = 1.16$) (Cohen's $d = 0.13$), while for the “originality” trait, women gave significantly higher scores ($M = 3.90$, $SD = 1.01$) than men ($M = 3.64$, $SD = 1.11$) (Cohen's $d = 0.25$).

6. Discussion

In the current research, we identified 47 traits which make flirting effective, and we classified them in nine broader categories. Having a good non-verbal behavior, being intelligent and having a gentle approach were rated as the most important factors. Sex difference were found for most of the factors. For instance, women rated non-verbal behaviors and gentle approach as more effective on them, while men rated good looks as more effective. Last but not least, older participants rated factors, such as the “Gentle approach,” to be more effective on them.

Traits, such as interest, intelligence, character, similarity and looks, are highly valued in the mating market (Buss, 2017); therefore, as predicted by our theoretical framework, they matched to traits which make flirting effective. More specifically, the “Non-verbal behaviors,” the “Courage and determination” and the “Strong interest and romance” indicate interest. In particular, an individual who has the courage to initiate flirting, is bold to make intense eye contact and persists even if the initial attempts do not work, indicates a strong interest for the target party. In the “Strong interest and romance” factor, traits such as “intense interest for me” and “interest to learn things about me” loaded, suggesting that participants interpreted intense interest for them as romantic. These factors signal potentially information about one's character: Flirting initiators are determined and persistent in achieving their goals.

In the “Intelligence” factor, flirting becomes effective by signaling intelligence presumably through correlates of it, such as education, being good with words and humor. The humor was by far the most frequently reported trait in Study 1, and had the second highest mean score in Study 2. It has been previously argued that it is a sign of intelligence (Howrigan & MacDonald, 2008; Miller, 2000). For instance, in one study, researchers employed a sample of students in the US, and tested them for verbal intelligence, abstract reasoning and their ability

to produce humor (Greengross & Miller, 2011). They found that high scores in intelligence correlated positively with abilities to recognize and produce humor (see also Feingold & Mazzella, 1993). Our finding that humor loaded to the “Intelligence” factor, suggests that participants viewed it also as a sign of intelligence. Nonetheless, its loading to this factor was only moderate, suggesting that humor was not viewed solely as a sign of intelligence.

The “Gentle approach,” the “Cheerfulness” and the “Strong interest and romance” are factors that convey information about character. More specifically, in the “Gentle approach,” traits, such as “seriousness,” “discreteness,” “maturity,” “good character” and “kindness” loaded. We can interpret this factor by assuming that the way the initiators of flirting approach their targets conveys information about their character. The “Cheerfulness” factor signaled information about character traits, such as “cheerfulness,” “easygoingness” and “positivity,” which make someone more pleasant to be with. Moreover, the “Strong interest and romance” factor conveys information about traits, such as tenderness and sweetness.

Furthermore, the “Good looks” conveys information about looks, while the “Common characteristics” conveys information that the two parties have things in common. Note that the “comfortable with his/her spending” trait loaded to the “Good looks,” suggesting that people assess how well-off individuals are by their looks (e.g., wearing expensive clothing). Finally, the “Mystery” factor most likely turns flirting effective, not because it conveys information about a trait which interests people, but because it captures their attention. Attention focuses on things which are out of the ordinary, so individuals who are out of the ordinary – have, for instance, unusual hobbies and originality in their approach – are more likely to catch one's attention. For example, an individual whose hobby involves travelling the world to collect rare flowers, may trigger the curiosity of a woman to learn more about him.

The hierarchy of factors partially reflects the hierarchy of mate preferences. For instance, research on mate preferences has found that character traits and intelligence are considered more important in a partner than looks and similarity (Buss, 2017), which can explain why factors, such as “Intelligence” and “Cheerfulness,” were ranked higher than factors, such as “Good looks” and “Common characteristics.”

Sex differences were found for most of the factors, indicating that the traits we have identified here do not have an equal effectiveness on each sex. More specifically, women ascribe more importance to the interest of a partner in them (Buss, 1989; 2017), so they rated factors, such as the “Courage and determination” and the “Strong interest and romance,” as more effective on them than men. As reflected in the “Gentle approach,” women ascribed much more importance to the approach of a partner. Actually, the effect size indicated that the largest sex difference was in this factor. One possible explanation is that women interpreted this factor as indicating a true interest in them. For instance, traits, such as “seriousness” and “maturity,” loaded here, so female participants may have thought that someone with such traits was looking for a long-term partner rather than a casual one. To put it differently, a rough approach could be interpreted to mean that the

flirting individual had intentions no other than sex. Another reason could be that women have been targets for sexual assault throughout human evolutionary history and thus, have evolved protection mechanisms (Fetchenhauer & Rohde, 2002). One such mechanism is likely to be ascribing considerable importance to the way someone approaches them, and avoiding the ones who do not have a gentle approach, i.e., who are rough and move too fast. The reason may be that such an approach could indicate sexual assault, if the woman does not give in to the flirting attempts.

Men place more importance to the looks of a prospective partner than women (Buss & Schmitt, 2019; Furnham, 2009), so they indicated that this trait would be more effective on them. On the other hand, women ascribe more importance to the resources that a partner controls (Buss & Schmitt, 2019), so they gave a higher importance to the “Comfortable with his/her spending” trait. Moreover, women tend to prefer intelligence in a partner more than men (Furnham, 2009), a possible reason being that this trait predicts resource generating capacity (Buss, 2017). Accordingly, women indicated that the “Intelligence” factor would be more effective on them than men.

Our findings were also consistent with the findings of previous studies. For instance, Wade and Feldman (2016) found that a strong demonstration of interest would make flirting effective, Walsh and Hewitt (1985) found that good eye contact and smile would make flirting effective, and White et al. (2018) found that being confident, smiley and demonstrate interest would make flirting effective. Yet, these studies examined individual traits and behaviors without classifying them into broader factors, preventing us from making direct comparisons.

Our research suggests that flirting which involves good eye contact, humor, pleasantness and a slow but persistent approach, as well as signs of intense interest, is likely to be quite effective. Its effectiveness will increase further, if it conveys information on similarity, is accompanied by good looks and an original way of approach. Furthermore, women's flirting on men would be more effective, if they have good looks, while men's flirting on women would be more effective, if they have a gentle approach, indicate strong interest and persistence, demonstrate signs of intelligence and wealth.

These findings could be used to develop interventions, which could enable people to improve the effectiveness of their flirting effort. Although it is true that things, such as personality and intelligence, cannot change with interventions, it is equally true that others can change. For instance, intervention could enable people to improve their way of approach, making it less rough and more gentle. Similarly, interventions could enable people to maintain good eye contact and improve their looks by dressing, for instance, better. Such interventions are important, because a considerable number of adult individuals face difficulties in attracting partners (Apostolou & Wang, 2019), with one of the main reasons being poor performance in flirting (Apostolou, 2017). Thus, they could be used by professional, such as counseling psychologists and flirt coaches, to enable their clients to improve their performance in the domain of mating.

The current research took place in Europe in a Western social context, so our findings are likely to apply to other similar contexts, such as the USA. They may not readily apply however, to cultural contexts, which are considerably different from the Western one. For instance, in poorer countries people may ascribe more importance to the resources a prospective mate controls; thus, signs of wealth such as “Comfortable with his/her spending” may turn flirting more effective than in the context of the present study. Furthermore, in certain Arab countries, such as Saudi Arabia, sexes are segregated and cannot flirt, so our findings are not relevant. Cross-cultural replication will enable a better understanding of how cultural variables affect the effectiveness of different flirting traits.

The current research is not without limitations. To begin with, our data are self-report and are thus, subject to biases, such as participants lacking a good understanding of which flirting traits would be more

effective on them. In addition, we employed a convenience sampling method, so our findings may not readily apply to the general population. Moreover, as discussed above, our research was confined to the Greek cultural context, so our results do not necessarily generalize to other cultural contexts. Furthermore, our study examined only sex and age effects, so it would be fruitful for future research to investigate the impact of other factors on the effectiveness of flirting. For instance, someone who scores high in the General Factor of Personality (GFP) is a good company (see van der Linden, te Nijenhuis & Bakker, 2010) and likely a good flirter is also good company. Accordingly, the association between and effectiveness in flirting needs to be investigated. Last but not least, people look for different traits in a casual than in a long-term partner (Buss & Schmitt, 2019). Thus, the effectiveness of the traits we have identified in the current study may differ with the type of partner a mating target is looking to find. Our research design did not distinguish between the two types of mating, an endeavor that future research should undertake.

In sum, flirting is crucial in mating and in the present research we have identified several traits that make it effective. We have also found that the effectiveness of these traits is contingent on the sex of the target. Much more research is necessary however, in order to better understand this fascinating phenomenon.

References

- Apostolou, M. (2017). Why people stay single: An evolutionary perspective. *Personality and Individual Differences*, 111, 263–271.
- Apostolou, M., & Panayiotou, R. (2019). The reasons that prevent people from cheating on their partners: An evolutionary account of the propensity not to cheat. *Personality and Individual Differences*, 146, 34–40.
- Apostolou, M., & Wang, Y. (2019). The association between mating performance, marital status, and the length of singlehood: Evidence from Greece and China. *Evolutionary Psychology*, 17(4), <https://doi.org/10.1177/1474704919887706>.
- Buss, D. M. (1989). Sex differences in human mate preferences: Evolutionary hypotheses tested in 37 cultures. *Behavioral and Brain Sciences*, 12, 1–14.
- Buss, D. M. (2017). *The evolution of desire: Strategies of human mating* (4th ed.). New York: Basic Books.
- Buss, D. M., & Barnes, M. (1986). Preferences in human mate selection. *Journal of Personality and Social Psychology*, 50, 559–570.
- Buss, D. M., & Schmitt, D. P. (2019). Mate preferences and their behavioral manifestations. *Annual Review of Psychology*, 70, 77–110.
- Clark, C. L., Shaver, P. R., & Abrahams, M. F. (1999). Strategic behaviors in romantic relationship initiation. *Personality and Social Psychology Bulletin*, 25, 709–722.
- Deshotels, T., & Forsyth, C. (2006). Strategic flirting and the emotional tab of exotic dancing. *Deviant Behavior*, 27, 223–241.
- Eibl-Eibesfeldt, I., & Hass, H. (1967). Film studies in human ethology. *Current Anthropology*, 8, 477–479.
- Feingold, A., & Mazzella, R. (1993). Preliminary validation of a multidimensional model of wittiness. *Journal of Personality*, 61, 439–456.
- Fetchenhauer, D., & Rohde, P. A. (2002). Evolutionary personality psychology and victimology: Sex differences in attitudes and short-term orientation and their relation to sex differences in victimizations. *Evolution and Human Behavior*, 23, 233–244.
- Furnham, A. (2009). Sex differences in mate selection preferences. *Personality and Individual Differences*, 47, 262–267.
- Grammer, K. (1990). Strangers meet: Laughter and nonverbal signs of interest in opposite-sex encounters. *Journal of Nonverbal Behavior*, 14, 209–236.
- Grammer, K., Kruck, K., & Juetten, A. (2000). Non-verbal behavior as courtship signals: The role of control and choice in selecting partners. *Evolution & Human Behavior*, 21, 371–390.
- Greengross, G., & Miller, G. (2011). Humor ability reveals intelligence, predicts mating success, and is higher in males. *Intelligence*, 39, 188–192.
- Gueguen, N. (2008). The effect of a woman's smile on men's courtship behavior. *Social Behavior and Personality: An International Journal*, 36, 1233–1236.
- Guerrero, L. K., Andersen, P. A., & Afifi, W. A. (2001). *Close encounters: Communicating in relationships*. Mountain View, CA: Mayfield.
- Henningsen, D. D. (2004). Flirting with meaning: An examination of miscommunication in flirting interactions. *Sex Roles*, 50, 481–489.
- Howrigan, D. P., & MacDonald, K. B. (2008). Humor as a mental fitness indicator. *Evolutionary Psychology*, 6, 652–666.
- Li, N. P., Valentine, K. A., & Patel, L. (2010). Mate preferences in the US and Singapore: A cross-cultural test of the mate preference priority model. *Personality and Individual Differences*, 50, 91–94.
- Loe, K. (1996). Working for men-At the intersection of power, gender, and sexuality. *Sociological Inquiry*, 66, 399–421.
- Luscombe, B. (2008, January). Why we flirt. *Time*, 171, 62–65.
- McCormick, N. B., & Jones, A. J. (1989). Gender differences in nonverbal flirtation. *Journal of Sex Education & Therapy*, 15, 271–282.
- Miller, G. F. (2000). Mental traits as fitness indicators: Expanding evolutionary

- psychology's adaptationism. *Annals of the New York Academy of Sciences*, 907, 62–74.
- Moore, M. (1985). Nonverbal courtship patterns in women: Context and consequences. *Ethology and Sociobiology*, 6, 237–247.
- Moore, M. (2010). Human nonverbal courtship behavior-A brief historical review. *Journal of Sex Research*, 47, 171–180.
- Moore, M., & Butler, D. (1989). Predictive aspects of nonverbal courtship behavior in women. *Semiotica*, 3, 205–215.
- Renninger, L., Wade, J., & Grammer, K. (2004). Getting that female glance: Patterns and consequences of male nonverbal behavior in courtship contexts. *Evolution and Human Behavior*, 25, 416–431.
- Schefflen, A. E. (1965). Quasi courtship behavior in psychotherapy. *Psychiatry: Journal for the Study of Interpersonal Processes*, 28, 245–257.
- Thomas, A. G., Jonason, P. K., Blackburn, J. D., Kennair, L. E. O., Lowe, R., Malouff, J., et al. (2019). Mate preference priorities in the East and West: A cross-cultural test of the mate preference priority model. *Journal of Personality Advance Online Publication*.
- van der Linden, D., te Nijenhuis, J., & Bakker, A. B. (2010). The general factor of personality: A meta-analysis of Big Five intercorrelations and a criterion-related validity study. *Journal of Research in Personality*, 44, 315–327.
- Wade, T. J., & Feldman, A. (2016). Sex and flirtation. *Human Ethology Bulletin*, 31, 30–44.
- Walsh, D. G., & Hewitt, J. (1985). Giving men the come-on: Effect of eye contact and smiling in a bar environment. *Perceptual and Motor Skills*, 61, 873–874.
- White, J., Lorenz, H., Perilloux, C., & Lee, A. (2018). Creative casanovas: Mating strategy predicts using—But not preferring—Atypical flirting tactics. *Evolutionary Psychological Science*, 4, 443–455.
- Willis, F., & Briggs, L. (1992). Relationship and touch in public settings. *Journal of Nonverbal Behavior*, 16, 55–63.
- Yelvington, K. A. (1996). Flirting in the factory. *Journal of the Royal Anthropological Institute*, 2, 313–333.